TEXORA · LECTURE SHEET

Managerial Economics

The manager's view of the market: demand, supply and elasticity, the cost concepts behind pricing, market structures from perfect competition to monopoly, and the marginal rule MR = MC that fixes the profit-maximising price and quantity.

TE-3109 · SUPPLY CHAIN MANAGEMENT IN TEXTILES · TOPIC 7 · LEVEL 3 TERM 1

1. What managerial economics does

Managerial economics uses economic theory and quantitative methods to solve practical business problems: what price to charge, how much to produce, which product mix to offer, whether to enter a new market, and how rivals will react. Unlike production economics (Topic 6), which looks *inside* the factory, managerial economics looks *outward* to buyers, competitors and the market — and then brings the two views together, because the right price depends on both what buyers will pay and what the product costs to make.

Its working tools are marginal analysis, demand estimation, elasticity, cost analysis and an understanding of market structure. Each is taken in turn, with textile and apparel examples throughout: a basic T-shirt sold to a mass retailer behaves very differently from a branded fashion garment, and the economics explains why.

2. Demand, supply and market equilibrium

Demand is the quantity buyers are willing and able to buy at each price in a given period. The **law of demand** states that, other things equal, quantity demanded rises as price falls — buyers substitute toward the cheaper good and new buyers enter. Demand depends also on buyers' incomes, prices of substitutes and complements, tastes and fashion, expectations and the number of buyers. A change in the good's own price moves the buyer *along* the demand curve; a change in any other factor *shifts* the whole curve — for example, a fashion trend shifts demand for a style outward at every price.

Supply is the quantity sellers offer at each price. The **law of supply** states that, other things equal, quantity supplied rises as price rises, because higher prices cover the rising marginal cost of extra output and draw in more capacity. Supply shifts with input prices (cotton, dyes, wages), technology, taxes or subsidies, and the number of sellers. Where demand and supply meet is the **market equilibrium**: the price at which quantity demanded equals quantity supplied. Above it, a surplus pushes price down; below it, a shortage pushes price up.

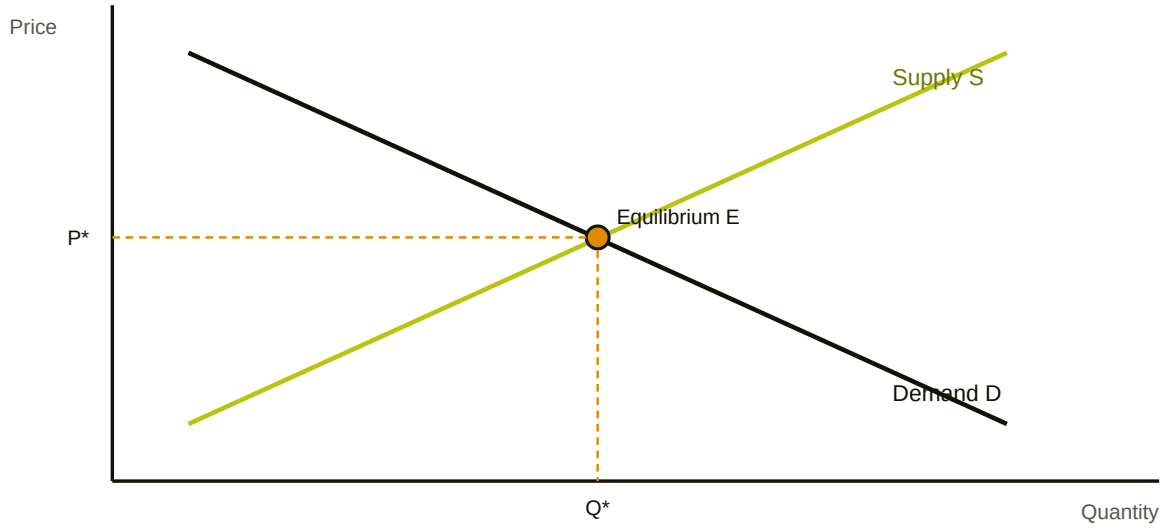


Figure 1 — Market equilibrium. Demand slopes down (law of demand) and supply slopes up (law of supply); they cross at price P* and quantity Q*. Any other price creates a surplus or a shortage that pushes the market back toward E.

3. Elasticity — how strongly buyers respond

Price elasticity of demand measures the percentage change in quantity demanded divided by the percentage change in price:

$$E_d = (\%\text{ change in quantity demanded}) \div (\%\text{ change in price})$$

Demand is **elastic** when $|E_d|$ is greater than 1 — quantity responds strongly, as with fashion garments that have many substitutes. It is **inelastic** when $|E_d|$ is less than 1 — quantity responds weakly, as with basic necessities or a specialised fabric only one mill can supply to specification. Elasticity decides whether a price rise helps or harms revenue: with elastic demand, a rise loses so many sales that revenue falls; with inelastic demand, revenue rises.

WORKED EXAMPLE — PRICE ELASTICITY AND REVENUE

A garment's price rises from Tk 500 to Tk 550, an increase of 10 %, and quantity sold falls from 20,000 to 17,000 pieces, a decrease of 15 %.

Price elasticity = −15 % ÷ 10 % = **−1.5**. The negative sign records the law of demand; the size, 1.5, means demand is **elastic**.

Revenue before = 500 × 20,000 = **Tk 1,00,00,000**. Revenue after = 550 × 17,000 = **Tk 93,50,000**. Revenue **falls** by Tk 6,50,000 — exactly what elastic demand predicts for a price rise. A price *cut* would have raised revenue for this product.

Related measures are used in planning. **Income elasticity** shows how demand moves with buyers' incomes — high for premium apparel, low or even negative for the cheapest basics. **Cross elasticity** shows how demand for one good responds to another's price: positive for substitutes (two competing T-shirt brands), negative for complements (shirts and matching trousers). A merchandiser forecasting next season's orders is, in effect, estimating these elasticities from past seasons' data.

4. Cost concepts for managerial decisions

Managers need cost classified for decisions, not only for accounts. **Total cost TC = TFC + TVC**, where TFC is total fixed cost and TVC is total variable cost. **Average cost AC = TC/Q** states the full cost per unit at a given output — the figure compared with price to judge profitability of the whole operation. **Marginal cost MC = ΔTC/ΔQ** states the cost of the *next* unit — the figure compared with marginal revenue to judge whether producing more is worthwhile. **Opportunity cost** counts the best alternative forgone; **sunk cost**, already spent, is excluded from forward decisions. Accounting profit deducts only explicit costs; **economic profit** also deducts opportunity costs, including the owner's capital and time — a factory can show an accounting profit yet earn no economic profit if its resources would earn as much elsewhere.

CONCEPT	FORMULA / MEANING	DECISION IT GOVERNS
Total cost (TC)	TFC + TVC	Overall viability, budgeting
Average cost (AC)	TC ÷ Q	Price vs full cost; break-even checks
Marginal cost (MC)	ΔTC ÷ ΔQ	Produce more or less; accept an extra order
Marginal revenue (MR)	ΔTR ÷ ΔQ	Sell more or less; set output where MR = MC
Opportunity cost	Best alternative forgone	Choice between orders, products, investments
Sunk cost	Past, unrecoverable spending	Excluded — never a reason to continue

5. Market structures

How much freedom a firm has over price depends on its market structure:

STRUCTURE	SELLERS / PRODUCT	PRICE POWER	TEXTILE ILLUSTRATION
Perfect competition	Many sellers, identical product	None — price taker	Grey fabric of a standard construction sold on spot markets (nearest real-world approximation)

Monopolistic competition	Many sellers, differentiated product	Some — brand, design, service	Branded apparel labels competing on style and brand image
Oligopoly	Few large sellers, interdependent	Considerable but watched — rivals react	Large dyeing-finishing groups or man-made fibre producers in a region
Monopoly	Single seller, no close substitute	Greatest — price maker, restrained by demand and regulation	A sole utility supplier of water or power to an industrial zone

Most Bangladesh garment exporters operate close to monopolistic competition among themselves but face powerful buyers — a near-oligopsony on the buying side — which compresses their price freedom. The strategic response visible across the industry follows directly from the theory: differentiate (design capability, compliance, speed, full-package service) to move away from pure price competition, because undifferentiated products earn only the market price.

6. Profit maximisation: MR = MC

A firm maximises profit where the revenue from one more unit exactly equals its cost: **MR = MC**. If MR exceeds MC, extra units add profit, so expand. If MC exceeds MR, the last units lose money, so contract. Price is then read off the demand curve at that quantity. Facing a downward-sloping demand curve, the firm must cut price to sell more, so MR lies below price and falls twice as fast for a straight-line demand curve.

WORKED EXAMPLE — PROFIT MAXIMISATION WITH MR = MC

A firm faces demand P = 1000 − 2Q (P in taka, Q in units). Marginal cost is constant at Tk 200, and fixed cost is Tk 20,000.

Total revenue TR = P × Q = 1000Q − 2Q², so marginal revenue MR = 1000 − 4Q.

Set MR = MC: 1000 − 4Q = 200, giving **Q = 200** units.

Price from demand: P = 1000 − 2(200) = **Tk 600**.

TR = 600 × 200 = **Tk 1,20,000**. TC = fixed cost + (MC × Q) = 20,000 + (200 × 200) = **Tk 60,000**.

Profit = TR − TC = 1,20,000 − 60,000 = **Tk 60,000** — the maximum available under this demand and cost structure.

Note what the rule does *not* say. It does not say charge the highest price — at Tk 800 only 100 units sell and profit is smaller. It does not say maximise output or revenue — revenue is highest where MR = 0, well past the profit peak. And a firm should shut down in the short run only if price falls below average variable cost, because any price above AVC still contributes something toward fixed costs. These

distinctions decide real cases: whether to run a line at a thin margin through a slack season, or to close it and lose trained workers.

7. Pricing and forecasting in practice

Textile pricing methods map onto the theory. **Cost-plus pricing** adds a margin to full cost — simple and safe for cost recovery, but blind to demand. **Target-return pricing** sets price to earn a chosen return on capital. **Value-based pricing** starts from the buyer's perception of value — the route brands take. **Competitor-based pricing** follows the market, unavoidable for undifferentiated basics. Mature managers combine them: cost sets the floor, competitors set the reference, value sets the ceiling, and elasticity predicts what a movement within that range will do to volume and revenue. Demand forecasting — trend analysis, buyer indications, and elasticity-adjusted price scenarios — then converts pricing choices into production and raw-material plans for the supply chain.

Glossary

Demand — quantities buyers will and can buy at each price. **Law of demand** — quantity demanded rises as price falls, other things equal. **Law of supply** — quantity supplied rises as price rises, other things equal. **Equilibrium** — price where quantity demanded equals quantity supplied. **Price elasticity of demand** — percentage change in quantity demanded per percentage change in price. **Marginal revenue** — extra revenue from one extra unit sold. **Economic profit** — revenue minus all costs including opportunity costs. **Oligopoly** — market dominated by a few interdependent sellers. **Monopolistic competition** — many sellers with differentiated products. **MR = MC rule** — profit is maximised where marginal revenue equals marginal cost.

Key points

• Demand slopes down and supply slopes up; their intersection is the market equilibrium, restored by surplus and shortage pressures.
• Elasticity measures responsiveness: in the worked example elasticity is −1.5 (elastic), so raising price from Tk 500 to Tk 550 cut revenue from Tk 1,00,00,000 to Tk 93,50,000.
• AC judges the whole operation; MC and MR judge the next unit. Sunk costs are ignored; opportunity costs are counted.
• Price power grows from perfect competition through monopolistic competition and oligopoly to monopoly; differentiation is how garment firms gain price power.
• Profit is maximised where MR = MC: in the worked example Q = 200, P = Tk 600, profit Tk 60,000.
• Sound pricing combines a cost floor, a competitor reference and a value ceiling, checked against elasticity.

Practice

1. Price falls from Tk 800 to Tk 720 and quantity rises from 10,000 to 11,000 units. Calculate the price elasticity of demand and state whether demand is elastic or inelastic. What happens to revenue?
2. A firm faces demand P = 800 − Q, with constant MC = Tk 200 and no fixed cost. Find the profit-maximising quantity, price and profit.
3. Distinguish a movement along a demand curve from a shift of the curve, giving one textile example of each.
4. Classify each market: (a) many boutiques selling differentiated dresses; (b) a single supplier of industrial water to an estate; (c) three large fibre producers watching each other's prices. Justify each answer briefly.
5. A factory's price is above AVC but below AC in a slack month. Using marginal reasoning, argue whether it should keep running in the short run.
6. Explain why MR lies below the demand (price) curve for a firm that must cut its price to sell more, and why this matters for the MR = MC calculation.

Answer key

1. Price change = −80 ÷ 800 = −10 %. Quantity change = +1,000 ÷ 10,000 = +10 %. Elasticity = 10 % ÷ −10 % = −1.0 — unit elastic. Revenue before = 800 × 10,000 = Tk 80,00,000; after = 720 × 11,000 = Tk 79,20,000. Strictly, revenue is almost unchanged — the small fall is an artefact of using end-point percentages; with unit elasticity revenue is essentially constant, as theory predicts.
2. TR = 800Q − Q^2, so MR = 800 − 2Q. Set MR = MC: 800 − 2Q = 200, so Q = 300. Price P = 800 − 300 = Tk 500. TR = 500 × 300 = Tk 1,50,000; TC = 200 × 300 = Tk 60,000; profit = Tk 90,000.
3. Movement along: the same T-shirt's price is cut and buyers purchase more of it — quantity changes, curve unmoved. Shift: a fashion trend makes the style popular, so more is demanded at every price — the whole curve moves outward. (Reverse examples — a cotton price rise shifting supply — are equally valid for supply.)
4. (a) Monopolistic competition — many sellers, differentiated products. (b) Monopoly — single seller, no close substitute. (c) Oligopoly — few sellers whose pricing decisions are interdependent.
5. Keep running in the short run. Price above AVC means every unit contributes (P − AVC) toward fixed costs. Fixed costs are incurred whether the line runs or not, so running loses less than closing. Close only if price falls below AVC, when each unit adds to the loss, or if the long-run outlook shows price persistently below AC with no recovery.
6. To sell one more unit the firm must lower the price on *all* units, not only the extra one. MR therefore equals the new price minus the revenue lost by cutting price on previous units, so MR is less than

price and falls faster. Using price in place of MR would overstate the gain from expansion and lead the firm to produce beyond the profit-maximising output.
